

MATH NATURAL LANGUAGE INFERENCE: THIS SHOULD BE EASY!

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ORIGINAL GOALS ON THIS PROJECT

THE MATH NLI PROBLEM

Are contemporary LLMs able to perform natural language inference (NLI) tasks on mathematical texts?

WHAT WE HAVE DONE, STARTING AROUND 2024

- ▶ corpus construction by hand: Our data can be found at <https://github.com/MathNLI/MathNLI>
- ▶ annotation by our team, including creation of guidelines
- ▶ use models as traditionally done
- ▶ also use models to write the hypotheses in NLI pairs
- ▶ ask new questions, especially about terms

OUR GROUP ORIGINATES FROM A “NALOMA-BASED” TEAM

Katerina Kalouli	Hai Hu	Alex Webb
LM	Valeria de Paiva	

This group focused on the SICK dataset and carried out a thorough examination and re-annotation of it.

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ORIGINAL MATHNLI TEAM

Quiye Gao	Hai Hu	Pavel Kovalev	Yikang Liu
LM	Valeria de Paiva	Zhiheng Qian	

CURRENT GROUP

Saul Blanco	Quiye Gao	Soren de Haan
LM	Valeria de Paiva	Daniel Quigley

Some of the work on this topic goes to conferences that you know, like *SEM.

In addition, there is a separate community grouped around meetings like

Conference on Intelligent Computer Mathematics (CICM).

CICM includes satellite events such as the MathUI Workshop.

Some work in the general computer/math area is a little farther afield from the language-oriented work here.

WE FIRST EXPECTED MATHNLI TO BE EASY

What counts as an entailment or contradiction in everyday-language texts is often taken as a complicated version of what happens with mathematics.

Facts, definitions, and logical reasoning play a large role in mathematical writing. Sentences ought to be precise and unambiguous.

So we might expect purported mathematical inferences to be **easier** to evaluate than those in everyday language.

And unlike language in the wild, the domain of mathematics is fairly well-defined.

THERE ARE COMPLICATIONS WITH MATHEMATICAL TEXT FROM THE START

The vocabulary may be unfamiliar to a generic audience: mathematical parlance can use daily words with new, unfamiliar meanings, e.g. 'ring', 'field', or even 'folklore'.

Visual elements such as symbols, equations, and diagrams, basically changes the very language of the text from plain text to a richer, multimodal language.

THE OVERALL EVALUATION

Despite historic performance on mathematical proofs, LLMs sometimes do not show a mastery of human-level causality, or of propositional reasoning and they don't always reason consistently in the first place.

Sometimes they deserve an A+ in advanced problem solving but a B in the basics.

EXAMPLES IN OUR HUMAN-CREATED CORPUS.

P (Premise)	H (Hypothesis)	Label
<i>A notion of central importance in categorical topology is that of topological functor.</i>	Topological functor is a notion of categorical topology.	E
<i>The problem of relating a factorization system to a pointed endofunctor is considered.</i>	The problem of relating a factorization system to a pointed endofunctor is not discussed.	C
<i>A notion of central importance in categorical topology is that of topological functor.</i>	There are many notions of central importance in categorical topology.	N

The Ps come from a set of abstracts of papers in one particular field, category theory.

We wrote the Hs

Abbr.	Model
GPT4	GPT-4
L2	Llama 2 70B
L3	Llama 3 70B
C3	Claude 3
Mistral	Mistral-large
L3.1	Llama-3.1-70B-Instruct
Q2	Qwen2-72B-Instruct
Mixtral	Mixtral-8x22B-Instruct-v0.1
DS	deepseek-llm-67b-chat
Ge2	gemma-2-27b-it

TERMINOLOGY

Top: Group 1: five initial LLMs;
 Bottom: Group 2: five later LLMs.

		GPT4	L2	L3	C3	Mistral	L3.1	Q2	Mixtral	DS	Ge2
C	p	82.9	90.5	70.9	91.8	79.8	88.8	87.3	75.4	92.7	85.1
	r	98.9	61.3	96.8	95.7	97.8	93.5	95.7	98.9	81.7	92.5
	f1	90.2	73.1	81.8	93.7	87.9	91.1	91.3	85.6	86.9	88.7
E	p	90.1	73.9	85.4	93.5	89.8	83.8	80.8	86.3	82.0	82.8
	r	97.8	88.2	94.6	92.5	84.9	89.2	90.3	88.2	78.5	82.8
	f1	93.8	80.4	89.8	93.0	87.3	86.5	85.3	87.2	80.2	82.8
N	p	95.5	56.2	91.8	87.8	81.8	81.7	84.7	85.5	67.6	75.3
	r	68.8	63.4	48.4	84.9	67.7	72.0	65.6	57.0	78.5	68.8
	f1	80.0	59.6	63.4	86.3	74.1	76.6	73.9	68.4	72.6	71.9
acc		88.5	71.0	79.9	91.0	83.5	84.9	83.9	81.4	79.6	81.4
avg	p	89.5	73.5	82.7	91.0	83.8	84.8	84.2	82.4	80.8	81.1
	r	88.5	71.0	79.9	91.0	83.5	84.9	83.9	81.4	79.6	81.4
	f1	88.0	71.0	78.3	91.0	83.1	84.7	83.5	80.4	79.9	81.1

IGNORING GIVEN INFORMATION

P: In the nilpotent case, this nerve is known to be a Kan complex.

H: This nerve is not known to be a Kan complex.

VAGUE PREDICATES

P: We worked through numerous examples to demonstrate the power of these notions.

H: We worked through two examples to demonstrate the power of these notions.

PROBLEMATIC BEHAVIOR OF LLMs

TEMPORAL REASONING

P: We take some first steps in providing a synthetic theory of distributions.

H: First steps in providing a synthetic theory of distributions date back to the previous century, and we will not reiterate them. **N**

Humans reason that if the first steps had been taken in the premise pair,
then the first steps could not have been taken in the last century, and they consider this a contradiction.

NAMING OF MATH ENTITIES

There is a problem with naming mathematical entities, e.g. “group B” vs. “group C” if this is only used as a generic name.

HOW GOOD ARE LLMs AT MATH NLI?

Next, we had the 4 models in Group 1 and 5 models in Group 2 label 89 chosen pairs.

The models in group 1 show unanimous agreement in 57 of the pairs (64%),

while the models in group 2 do so in 65 (73%).

In group 1, for 50 of these 65 pairs (87%), their unanimous label agrees with human labels;

while the agreement for group 2 is 57 pairs (88%).

HERE IS OUR CONCLUSION

If we take the unanimous labels from the group 2 models to simply **be** the gold label, then this label is the same as the human label 88% of the time.

ASKING CHATGPT TO WRITE HYPOTHESES (WITH LABELS)

The overall accuracy of LLMs varies between 59.6% and 86.5%, which is **relatively lower than the accuracy on the seed corpus**.

The E and C pairs generated by GPT show a certain level of consistency relative to our seed pairs,
N pairs are frequently misclassified as E.

Surprisingly, Llama 2 classifies 75.9% of N pairs as E.

Among all evaluated models, Mixtral showed the least susceptibility to this issue, maintaining the highest accuracy of 76.0%.

Although its performance on the seed corpus was not outstanding, Mixtral achieved the highest overall accuracy of 86.5% on the GPT-generated corpus.

Furthermore, it was observed that
LLMs tend to identify C pairs within the GPT-generated corpus more accurately than they do within the seed corpus;
8 out of 9 models achieved an F1 score of over 90%.

Notably, Gemma2 successfully detected all the C samples in the GPT-generated corpus.

CONCLUSION OF THIS PART OF THE WORK

Contrary to what we initially assumed

Math NLI is not much easier than ordinary NLI for LLMs.

CHALLENGES

- ▶ unfamiliar vocabulary (e.g., 'ring', 'field', 'comonad'),
- ▶ multimodal elements like symbols and equations,
- ▶ lack of open-source mathematical resources,
- ▶ difficulty of finding expert human annotators.

LLMs show paradoxical performance on math tasks.

Their reasoning sometimes shows flaws.

But post-GPT LLMs avoid some issues that plagued earlier systems. For example, lexical ambiguity is no longer an issue.

("stack" vs. "stack")

SECOND LINE OF WORK: MATHEMATICAL TERMINOLOGY

This is a topic where ordinary NLI and MathNLI diverge.

We are finding it to be challenging,
and while we have some leads,
we are not sure that we have a satisfactory solution.